Spacecraft Attitude Control using Reaction Wheels

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Outline

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2. Dynamics
3. Simulation Setup
4. Results
5. Gain Optimization
6. Comparison
7. Conclusion

What is Spacecraft Attitude Control?

- **Attitude** refers to the orientation of a spacecraft in space—its angular position relative to a reference frame (e.g., Earth, stars, or orbit path).
- **Attitude control** is the process of actively adjusting or maintaining this orientation using actuators such as:
  - Reaction wheels
  - Control moment gyroscopes
  - Magnetorquers
  - Thrusters
- Controlled using feedback systems to reject disturbances and ensure precision.

Why Is it Important?

- Precise orientation is vital for:
  - Imaging missions
  - Communications
  - Orbital maneuvers
- Reaction wheels use angular momentum conservation to control attitude
- Enable high-precision line-of-sight stability without fuel

Project Objectives

- Simulate spacecraft attitude control in Python
- Implement PD controller with saturation and torque limits
- Design for performance:
  - Settling time < 5 s
  - Overshoot $< 2^\circ$
- Compare performance across spacecraft sizes

Governing Dynamics

Reaction Wheel Torque

$$\tau = I_{rw}\dot{\omega}$$

Control Law (PD)

$$\tau = -k_1 e - k_2 \dot{e}$$

Closed Loop

$$I_{sc}\ddot{\theta} + k_2\dot{\theta} + k_1\theta = 0$$

Dynamics - State Space Equations

State-Space Equation

$$\dot{x}(t) = Ax(t) + Bu(t) \tag{1}$$
$$y(t) = Cx(t) + Du(t) \tag{2}$$

States

$$\mathbf{x}(t) = \begin{bmatrix} \theta(t) \\ \omega_{sc}(t) \end{bmatrix} \tag{3}$$

Command Torque

$$\tau_{cmd}(t) = -k_1\theta(t) - k_2\omega_{sc}(t) \tag{4}$$

Dynamics - State Space Equations

Kinematic Equation

$$\dot{\theta}(t) = \omega_{sc}(t) \tag{5}$$

Dynamic Equation

$$\dot{\omega}_{sc}(t) = -\frac{k_1}{I_{sc}}\theta(t) - \frac{k_2}{I_{sc}}\omega_{sc}(t) \tag{6}$$

Characteristic Equation - Setup

Characteristic Equation

$$\det(sI - A) = s^2 + \frac{k_2}{I_{sc}} s + \frac{k_1}{I_{sc}} = 0 \tag{7}$$

Test Cases

Large Spacecraft (3 cases):

- Inertia: 1,000 $kg \cdot m^2$
- Wheel: 1–20 $kg \cdot m^2$
- Max speed: 3,000–25,000 RPM
- Max torque: 2–50 Nm

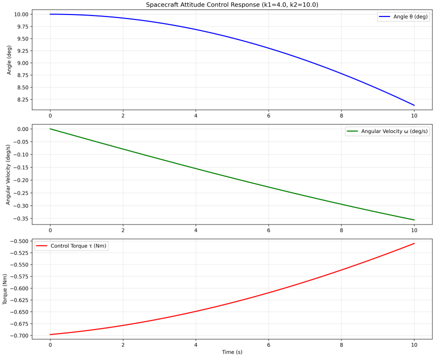
Small Spacecraft (3 cases):

- Inertia: 10 $kg \cdot m^2$
- Wheel: 0.05–0.5 $kg \cdot m^2$
- Max speed: 500–30,000 RPM
- Max torque: 0.1–1 Nm

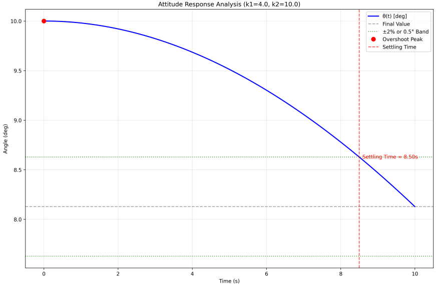
Test Conditions:

- 10° step disturbance
- Requirements: $T_s < 5$ s, overshoot $< 2^\circ$
- Full parameter space sweep

Large Spacecraft Results - Time Domain & Analysis



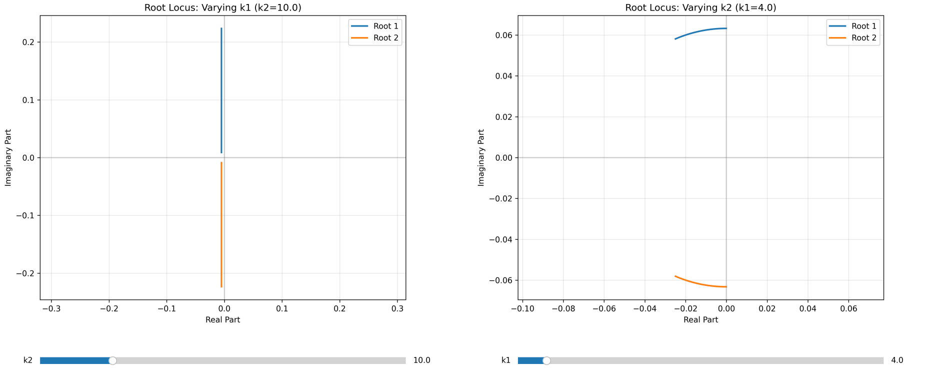
(a) Time-domain graphs



(b) Response analysis graph

Figure: Time-domain and Analysis of manually input gains for small spacecraft case 1

Large Spacecraft Results - Root Locus

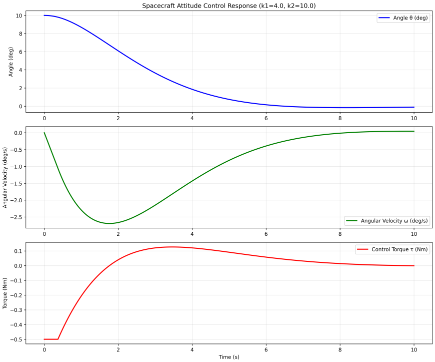


(a) Fixed k_2

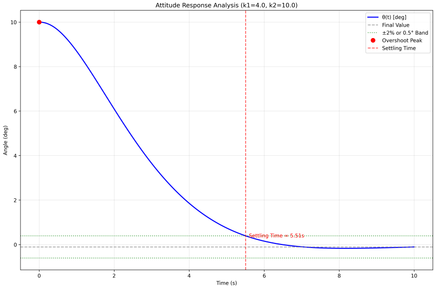
(b) Fixed k_1

Figure: Root Locus plots of manually input gains for large spacecraft case 1

Small Spacecraft Results - Time Domain & Analysis



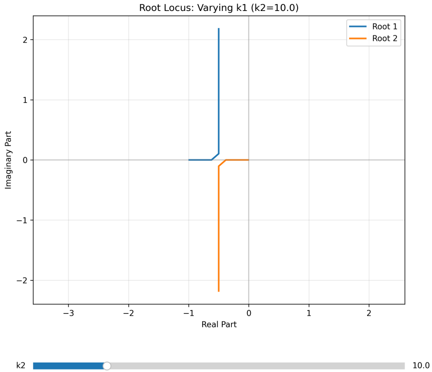
(a) Time-domain graphs



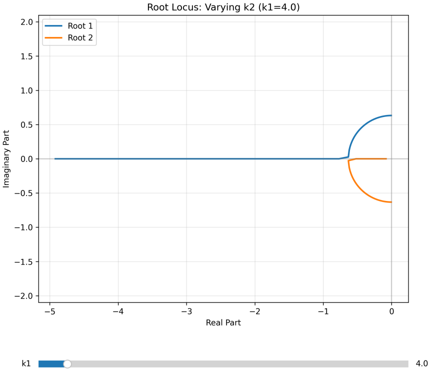
(b) Response analysis graph

Figure: Time-domain and Analysis of manually input gains for small spacecraft case 1

Small Spacecraft Results - Root Locus



(a) Fixed k_1



(b) Fixed k_2

Figure: Root Locus plots of manually input gains for small spacecraft case 1

Gain Optimization Function

- Estimate the most optimal range of gains
- Test gain combinations the resulting behavior
- Compare created gain combinations

Gain Optimization Cost Function

- Cost = weighted sum of:
  - Settling time
  - Overshoot
  - Torque usage
- Grid search across (k_1, k_2) domain
- Selected optimal gains minimizing cost

Gain Optimization Issues

- Failed to find valid optimal gains
- Valid optimal gains found were skewed to the extreme

Gain Optimization Solution

- Created the estimate gains function
  - Produced a range of possible gain based on the spacecraft dimensions

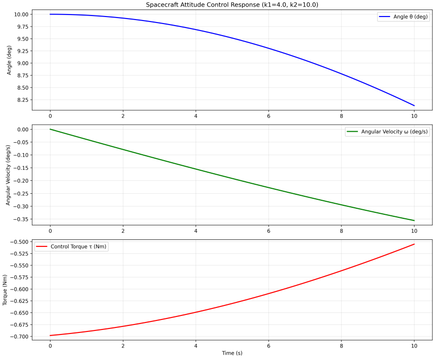
Program Inefficiency

- Uniform sampling: $200 \times 200 = 40{,}000$ combinations
- Ignores sensitivity variations across domain
- Inefficient in flat or highly sensitive regions

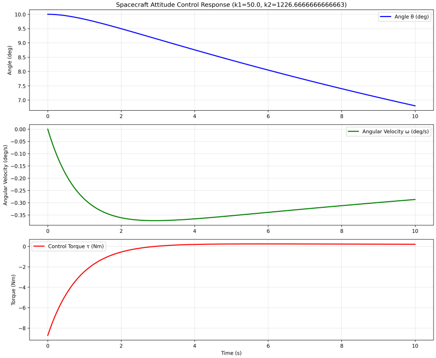
Program Inefficiency - Solution

- Gain estimation function reduced necessary values to loop over
- Reduced resolution for search from $100 \longrightarrow 20$
- Uniform sampling: $20 \times 20 = 4{,}00$ combinations

Large Spacecraft Results Comparison - Time-Domain



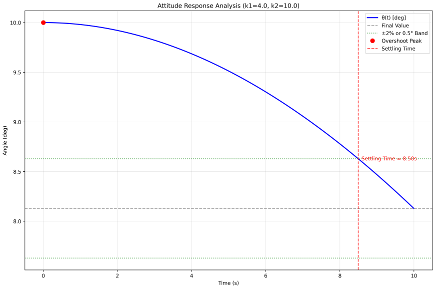
(a) Manually Input Gains



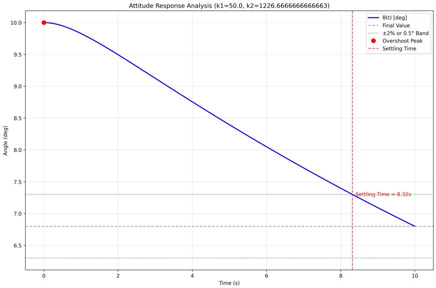
(b) Optomized Gains

Figure: Time-domain graphs for large spacecraft case 1

Large Spacecraft Results Comparison - Analysis



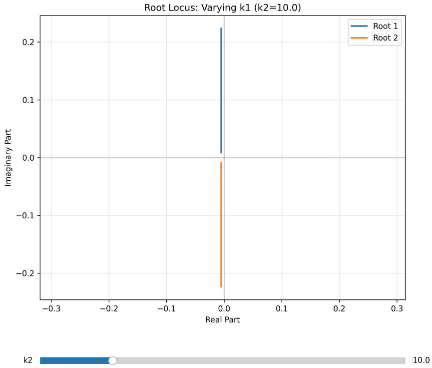
(a) Manually Input Gains



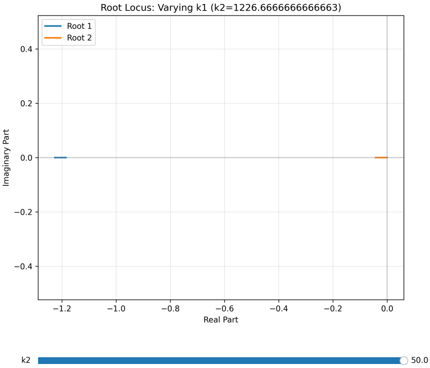
(b) Optimized Gains

Figure: Response Analysis graphs for large spacecraft case 1

Large Spacecraft Results Comparison - Root Locus(k_1)



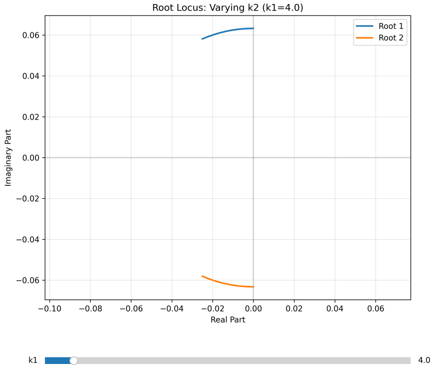
(a) Manually input gains



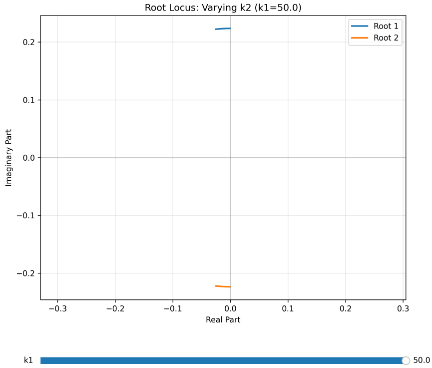
(b) Optimized gains

Figure: Root locus plots of a varying k_1 for large spacecraft case 1

Large Spacecraft Results Comparison - Root Locus(k_2)



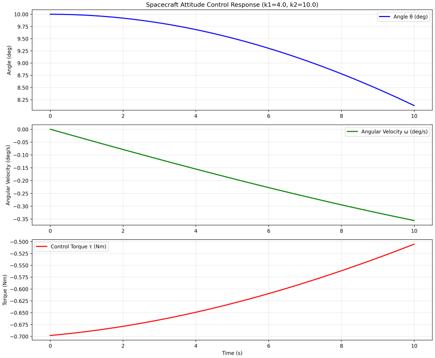
(a) Manually input gains



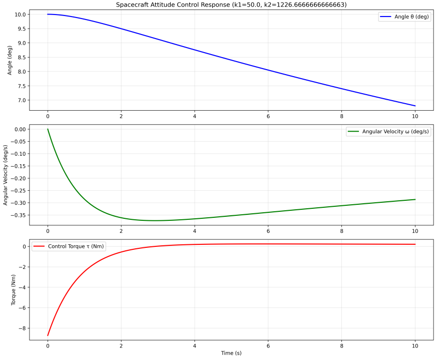
(b) Optimized gains

Figure: Root locus plots of a varying k_2 for large spacecraft case 1

Small Spacecraft Results Comparison - Time-Domain



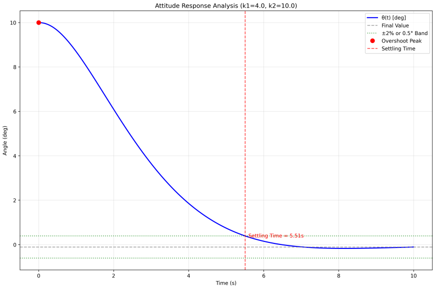
(a) Manually input gains



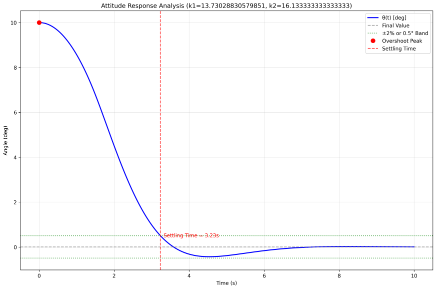
(b) Optimized gains

Figure: Time-domain graphs for small spacecraft case 1

Small Spacecraft Results Comparison - Response Analysis



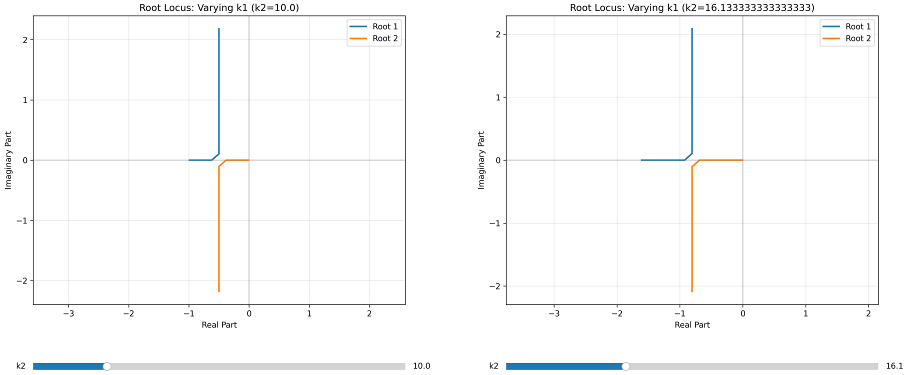
(a) Manually input gains



(b) Optimized gains

Figure: Response analysis graphs for small spacecraft case 1

Large Spacecraft Results Comparison - Root Locus (k_1)

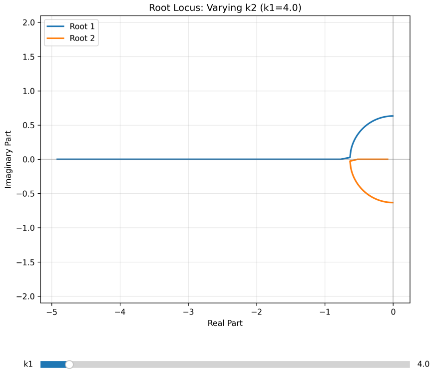


(a) Manually input gains

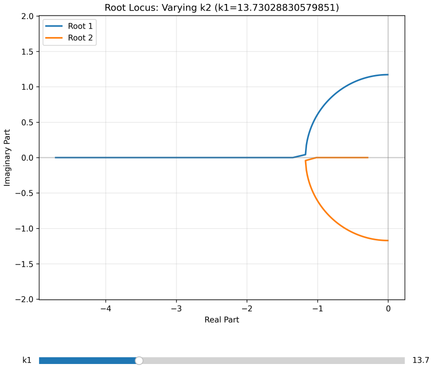
(b) Optimized gains

Figure: Root locus plots for a varying k_1 of small spacecraft case 1

Large Spacecraft Results Comparison - Root Locus (k_2)



(a) Manually input gains



(b) Optimized gains

Figure: Root locus plots for a varying k_2 of small spacecraft case 1

Summary of Findings

- Optimized PD gains significantly improve performance
- Settling time improved by 42%, torque reduced by $10\times$
- Simulation validated across small and large spacecraft

Future Work

- Implement nonlinear controllers (e.g. sliding mode, LQR)
- Extend to 3-axis attitude control
- Use intelligent gain tuning (e.g. Bayesian optimization)

Thank You!

Any Questions?